

Faculty of Applied Sciences - Rajarata University of Sri Lanka
COM 1302 – Database Management Systems
Practical Sheet

Practice 1

1. Connecting to Server,

mysql -h DatabaseHost -u DatabaseUserName -p DatabasePassword

Eg:

databaseUserName -> studentusername

Databasepassword -> dbpwd

databaseHost -> 127.16.17.253

mysql -h 127.16.17.253 -u studentusername -p dbpwd

In Local Server,

mysql -u root -p

2. Selecting a Database,

Create Database -> CREATE DATABASE databasename;

View Databases -> SHOW DATABASES;

Select a Database -> USE databasename;

Delete a Database -> DROP DATABASE databasename;

3. Creating Table,

Create a Table -> CREATE TABLE tablename(
column1 datatype,
column2 datatype,
column3 datatype,
.....
columnN datatype,
PRIMARY KEY(one or more columns)
);

View Tables -> SHOW TABLES;

View table information -> DESC / DESCRIBE tablename;

Delete a Table -> DROP TABLE tablename;

4. Populating Table,

Insert data to table -> INSERT INTO tablename
(column1, column2, column3,...columnN)
VALUES (value1, value2, value3,...valueN);

INSERT INTO tablename VALUES

(value1,value2,value3,...valueN);

• Create below Customers table,

ID	Name	Age	Address	Salary
1	Ramesh	32	Negombo	20000.00
2	Kumara	25	Colombo	15000.00
3	Kaushika	23	Jaffna	20000.00
4	Chethali	25	Perdeniya	58000.00
5	Harshana	27	Kurunegala	35000.00
6	Ashani	22	Kandy	45000.00
7	Nimali	24	Gampaha	10000.00

Table 1 Customers Table

5. Drop Table,

Delete a Table -> DROP TABLE tablename;

Delete only the Data Structure -> TRUNCATE TABLE tablename;

6. Selecting Fields,

Get all the Fields -> SELECT * FROM tablename;

Get specific Columns -> SELECT column1, column2, columnN
FROM tablename;

WHERE,

SELECT column1, column2, columnN
FROM tablename
WHERE [condition];

Operator	Description
=	Equal
>	Greater than
<	Less than
>=	Greater than or equal
<=	Less than or equal
<>	Not equal. Note: In some versions of SQL this operator may be written as !=
BETWEEN	Between a certain range
LIKE	Search for a pattern
IN	To specify multiple possible values for a column

Practice 2

Now, we will create the following table in the EMPLOYEE database.

```
CREATE TABLE EMPLOYEE(Fname VARCHAR(15),  
Minit CHAR,  
Lname VARCHAR(15),  
Ssn CHAR(9),  
Bdate DATE,  
Address VARCHAR(15),  
Sex CHAR,  
Salary DECIMAL(10,2),  
Super_ssn CHAR(9),  
Dno INT);
```

1. Get a Description of the Tables

- We use the DESCRIBE command to show the structure of our table, such as column names, constraints etc.
- The DESC command is a short form of the DESCRIBE command.

```
DESCRIBE employee;  
DESC employee;
```

2. Modify the Tables

MySQL allows you to modify tables. The following commands will help you with them.

Add new columns

```
ALTER TABLE EMPLOYEE  
ADD tel_no VARCHAR (255);
```

Drop columns

```
ALTER TABLE EMPLOYEE  
DROP COLUMN tel_no; Modify Columns  
ALTER TABLE EMPLOYEE  
MODIFY COLUMN Address VARCHAR(30);
```

3. Insert Data to the Table

There are two methods to insert data to the table.

Method 1

```
INSERT INTO EMPLOYEE  
(Fname, Minit, Lname, Ssn, Bdate, Address, Sex, Salary, Super_ssn, Dno)  
VALUES  
('John', 'B', 'Smith', '12345789', '1965-01-09', '731 Fondren, Houston, TX', 'M', 30000, '33344555', 5);
```

Method 2 - Insert Multiple Records

```
INSERT INTO EMPLOYEE(  
Fname, Minit, Lname, Ssn, Bdate, Address, Sex, Salary, Super_ssn, Dno)  
VALUES  
('Franklin', 'T', 'Wong', '33344555', '1955-12-08', '638 Voss, TX', 'M', 40000, '88866555', 5),  
('Alicia', 'J', 'Zelaya', '999887777', '1968-01-19', '3321 Castle, Houston, TX', 'F', 25000, '987654321', 4);
```

4. Update Tables

There are several ways in which you can update a table.

Update a single record

```
UPDATE EMPLOYEE  
SET Fname = 'Nimal'  
WHERE Ssn = '33344555'; Update Multiple Records  
UPDATE EMPLOYEE  
SET salary = 50000.00  
WHERE Dno=5;
```

Update Multiple Columns

```
UPDATE EMPLOYEE  
SET salary = 45000.00, Dno='2'  
WHERE Super_ssn='88866555';
```

Be careful when updating records. If you omit the WHERE clause, ALL records will be updated!

Delete

You can delete data in tables in the following methods.

Delete Specific Records

DELETE FROM EMPLOYEE WHERE Fname = 'Alicia';

Delete All Records

DELETE FROM EMPLOYEE;

Drop a Database

DROP DATABASE COMPANY;